

BINLEVEL · REQUIREMENTS

BinLevel requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-25	Amish Chadha	Initial scaffold

BinLevel requirements

These are first-pass requirements for the concept. Targets are proposals awaiting Amish and co-design partners, and will be checked by calculation at TRL 3. Status reflects the TRL 2 concept estimates in the design precis (BNL-PRC-001). Three requirements are not met or are at risk (R2, R5 and R10); see the notes under Table 1.

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status
R1	Measure distance from lid to waste surface over the full depth of the target container	0.03 to 1.5 m below the sensor face	Datasheet review; later bench test in a container	Met by design with two sensors (ultrasonic 0.25 to 1.5 m, ToF 0.03 to 0.4 m, estimates)
R2	Report fill level accurately on real mixed waste	Within ± 10 % of container depth for 90 % of readings	Field comparison against manual dip readings	At risk: bags, cardboard and uneven surfaces scatter the echo; unverified
R3	Report often enough to plan routes, and quickly when full	Routine uplink at least every 2 h; threshold alert within 20 min of crossing the set fill level (default 80 %)	Timing calculation; firmware configuration	Met by design: ranging every 15 min
R4	Run for years on one battery	5 years or more in the worst radio case (SF12); 10 years design target	Power budget calculation	Met on paper: about 18 years (SF12, C cell) estimate; enclosure and seals, not the cell, limit life
R5	Survive rain, condensation and washing	IP67 minimum; IP69K target where containers are pressure washed	Enclosure datasheet; later spray test	Not met for IP69K: stock IP67 box proposed
R6	Operate across the climate range	-20 to +60 °C inside the container	Datasheet review	Met on paper for cell and radio; transducer and ToF window condensation unverified
R7	Survive lifting, tipping and lid slams	Repeated shocks of the emptying cycle without loosening; target IK08 impact on the enclosure	Design review; later drop and shake test	Unverified; bracket with four through-bolts proposed
R8	Detect and report emptying and lid events	Tip event from the accelerometer reported with the next uplink	Firmware logic review	Met by design
R9	Warn of abnormal heat in the bin	Alert uplink within 15 min when temperature exceeds 70 °C or rises more than 15 K in 15 min	Firmware logic review	Met by design; maintenance aid only, not a fire detector
R10	Connect over open LoRaWAN within radio rules	LoRaWAN 1.0.x Class A; EU868, US915 and IN865 plans; within duty cycle and TTN's 30 s per day fair use	Airtime calculation; later range test	At risk in steel containers: internal antenna shielded; external antenna option needed
R11	Protect privacy	No camera or microphone; payload limited to fill, distance, temperature, tilt, events and battery	Design review of BOM and payload	Met by design
R12	Install without wiring or tools beyond a drill	Fit to a lid in 10 min or less with four bolts or rivets; no cables	Timed trial at TRL 4	Met by design (estimate)

ID	Requirement	Target	Verification (TRL 3 or later)	Concept status
R13	Resist casual tampering and theft	Mounted inside the lid; tamper-resistant fasteners	Design review	Met by design
R14	Low cost and buildable	Parts cost \$60 or less per sensor at prototype quantities; no custom tooling	Priced BOM	Met: about \$55 (indicative)
R15	Open and interoperable	Documented payload format and open decoder; works with any LoRaWAN network server	Documentation review	Met by design
R16	Light and compact	400 g or less; no larger than 160 x 90 x 100 mm below the lid, including bracket	Massing model, then weighing	Met: about 250 g, 150 x 80 x 63 mm below the lid (estimates)

Notes on requirements not met or at risk:

- **R2 at risk.** Ultrasonic echoes from bags, boxes and bulky items are uneven, and the waste surface is rarely flat. The mitigation is to take the median of several pings and to combine the two sensors, but accuracy on real waste is unknown until measured.
- **R5 not met for IP69K.** A low-cost stock enclosure is typically rated IP67. Where containers are washed with hot pressure water, a sealed potted design or an IP69K enclosure is needed, which raises cost.
- **R10 at risk.** An antenna inside a steel container and under a steel lid will lose much of its signal. Plastic (HDPE) containers are fine. Steel containers need an antenna fitted to the outside of the lid or rim.

■ Assumptions

- Target container: 1,100 L four-wheel communal container, internal depth about 1.07 m (estimate from the massing model).
- Fill level is reported as a percentage of the distance from the sensor face to the container floor, measured at installation.
- Worst radio case is spreading factor 12 at 125 kHz with a 12-byte payload (about 1.5 s airtime per uplink, estimate); typical urban case is SF9 (about 0.21 s).
- Temperatures inside a closed dark lid may reach about 60 °C in summer (estimate, to be measured).